

BUILD DOCUMENTATION

Cisco Catalyst 2960-X — SUE-SW1

Date: September 2026

Status: Configured / Operational

01 — OVERVIEW

The Cisco Catalyst 2960-X serves as the primary network switch for SUE LAB.

The switch provides a foundation for hands-on networking and cybersecurity experimentation, including VLANs, access ports, network segmentation, switch security, and Cisco IOS configuration.

02 — SPECIFICATIONS

- **Manufacturer:** Cisco
- **Model:** Catalyst 2960-X
- **Hostname:** SUE-SW1
- **Management Interface:** VLAN 1
- **Lab VLAN:** VLAN 10 — SUE-LAB
- **Configured Access Ports:** Gi1/0/1, Gi1/0/2
- **Console Speed:** 9600 baud
- **Rack:** 12U Open-Frame Rack

Key Features

- Managed Ethernet switching
- VLAN support
- Cisco IOS
- Console management
- Gigabit Ethernet connectivity
- Network segmentation
- Access and trunk port configuration
- Switch security features

03 — WHY I CHOSE IT

The Cisco Catalyst 2960-X was selected as the primary switch for SUE LAB to provide hands-on experience with enterprise networking equipment.

It provides a platform for practicing:

- VLAN configuration
 - Access ports
 - Trunk ports
 - Network segmentation
 - Switch management
 - Port security
 - BPDU Guard
 - Unused-port security
 - Network troubleshooting
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04 — INSTALLATION / CONFIGURATION

The Cisco Catalyst 2960-X was installed in the SUE LAB rack and accessed through its physical console interface.

Basic Configuration

The switch was configured with the hostname:

SUE-SW1

Basic switch security was configured, including an encrypted enable secret and password encryption.

Management Interface

VLAN 1 was configured as the initial management interface.

A private management IP address was assigned to the interface along with a default gateway.

Security Note: Private IP addresses and authentication credentials are intentionally not published in this documentation.

SUE LAB VLAN

A dedicated lab VLAN was created:

VLAN 10

Name: SUE-LAB

Lab Access Ports

Two Gigabit Ethernet ports were configured as access ports and assigned to VLAN 10:

Gi1/0/1 → VLAN 10

Gi1/0/2 → VLAN 10

Verification

The configuration was verified using Cisco IOS commands including:

- show ip interface brief
- show interfaces status
- show vlan brief
- show running-config

The configuration was saved to startup configuration to ensure it persists after reboot.







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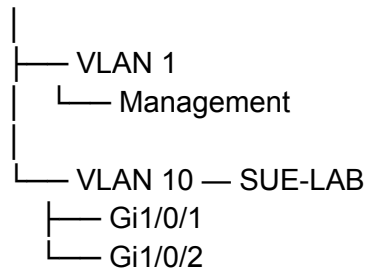
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05 — FINAL SETUP

The Cisco Catalyst 2960-X is installed as **SUE-SW1** and currently provides the switching foundation for SUE LAB.

Current VLAN Layout

SUE-SW1



The switch configuration has been saved to startup configuration, and console access has been successfully established.

The lab network will be expanded as additional networking and security components are added.

User Access Verification

Password:
SUE-SW1>show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gi1/0/3, Gi1/0/4, Gi1/0/5 Gi1/0/6, Gi1/0/7, Gi1/0/8 Gi1/0/9, Gi1/0/10, Gi1/0/11 Gi1/0/12, Gi1/0/13, Gi1/0/14 Gi1/0/15, Gi1/0/16, Gi1/0/17 Gi1/0/18, Gi1/0/19, Gi1/0/20 Gi1/0/21, Gi1/0/22, Gi1/0/23 Gi1/0/24, Gi1/0/25, Gi1/0/26 Gi1/0/27, Gi1/0/28, Gi1/0/29 Gi1/0/30, Gi1/0/31, Gi1/0/32 Gi1/0/33, Gi1/0/34, Gi1/0/35 Gi1/0/36, Gi1/0/37, Gi1/0/38 Gi1/0/39, Gi1/0/40, Gi1/0/41 Gi1/0/42, Gi1/0/43, Gi1/0/44 Gi1/0/45, Gi1/0/46, Gi1/0/47 Gi1/0/48, Gi1/0/49, Gi1/0/50 Gi1/0/51, Gi1/0/52
10	SUE-LAB	active	Gi1/0/1, Gi1/0/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
--More--			

SUE-SW1>show interface status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi1/0/1		notconnect	10	auto	auto	10/100/1000BaseTX
Gi1/0/2		notconnect	10	auto	auto	10/100/1000BaseTX
Gi1/0/3		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/4		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/5		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/6		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/7		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/8		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/9		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/10		notconnect	1	auto	auto	10/100/1000BaseTX
Gi1/0/11		notconnect	1	auto	auto	10/100/1000BaseTX

SUE-SW1>show version
Cisco IOS Software, C2960X Software (C2960X-UNIVERSALK9-M), Version 15.0(2)EX5, RELEASE SOFTWARE (fc1)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2014 by Cisco Systems, Inc.
Compiled Fri 21-Feb-14 05:54 by prod_rel_team

ROM: Bootstrap program is C2960X boot loader
BOOTLDR: C2960X Boot Loader (C2960X-HBOOT-M) Version 15.2(2r)E1, RELEASE SOFTWARE (fc1)

SUE-SW1 uptime is 53 minutes
System returned to ROM by power-on
System restarted at 10:58:14 UTC Sun Sep 13 2026
System image file is "flash:/c2960x-universalk9-mz.150-2.EX5/c2960x-universalk9-mz.150-2.EX5.bin"

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06 — CHALLENGES

Console Access

Connecting to enterprise networking equipment required establishing a serial console connection and configuring the appropriate terminal settings.

Cisco IOS

The initial configuration required learning Cisco IOS command modes and understanding the difference between user EXEC, privileged EXEC, and global configuration modes.

VLAN Configuration

Creating a VLAN and assigning ports to that VLAN are separate configuration steps.

The VLAN must first be created before physical switch ports can be assigned to it.

07 — WHAT I LEARNED

This build provided hands-on experience configuring a physical Cisco network switch.

Key Lessons

- Cisco IOS command-line configuration
- Switch management
- VLAN creation
- Access-port configuration
- Network segmentation
- Interface verification
- Configuration troubleshooting
- Saving switch configurations
- Basic switch security
- Enterprise networking concepts

The switch provides a physical platform for applying networking concepts in a controlled lab environment.

08 — NEXT STEPS

1. Secure unused switch ports
2. Create additional lab VLANs
3. Configure additional access ports
4. Configure trunk ports
5. Implement port security
6. Configure BPDU Guard
7. Disable unused interfaces
8. Add and configure a firewall/router to establish the SUE LAB network.
9. Implement DHCP and routing
10. Connect additional lab devices
11. Document the completed SUE LAB network